

REU-HPC: Week-2 Group Meeting

Group Meetings

Tuesday 3-4pm (starting June 16)

<https://utexas.zoom.us/j/85978334209>

Weekly Progress Report

Beginning next week, please send a brief **half page progress report** before each group meeting. The report does not need to be formal. Bullet points are perfectly acceptable.

Suggested contents:

What I worked on this week

Tasks completed, software installed, documentation reviewed, experiments performed, ...

What I learned

New HPC concepts, tools or techniques learned, technical challenges encountered, ...

Questions / Issues

Problems that need discussion, topics requiring clarification

The purpose is to help track progress and identify areas where additional guidance may be helpful.

Week 2 and 3 Objectives (June 8 – June 19)

The primary goal for week 2 and 3 is to become familiar with:

- TACC systems and basic HPC workflows
You should now be able to submit and run jobs on Frontera, Stampede3, Lonestar6, and Vista using allocations: **EAR23006 (for Frontera)** and **ASC26067 (Stampede3, Lonestar6, and Vista)**.
Please try the case `"/work2/05392/cylu/share/reu_2026/1.training/1.scaling_test"` on Frontera first.
- PEAK profiler and software/package installation
Get familiar with the PEAK profiler, including its basic workflow and usage. Also learn how to install software packages and dependencies on TACC systems.

REU Research Plan: Profiling Math Libraries in Scientific and AI/ML Applications Using PEAK

Project Overview

The goal of this eight-week REU project is to help you gain hands-on experience with performance profiling on high-performance computing systems and to apply profiling techniques to real scientific and AI/ML applications. The project will focus on using **PEAK**, a function-level profiling tool, to understand how important math libraries are used in practical computational workloads.

Many scientific and AI/ML applications rely heavily on optimized math libraries such as **BLAS**, **LAPACK**, and **FFTW**. These libraries provide core computational kernels for dense and sparse linear algebra, eigenvalue problems, matrix factorizations, Fourier transforms, and other operations that are widely used in simulation, data analysis, and machine learning workflows. Understanding how frequently these libraries are used, how much runtime they contribute, and how their usage patterns vary across applications is important for evaluating application performance on HPC systems.

This project has two major research directions:

1. **Math libraries in scientific applications**
2. **Math libraries in AI/ML workflows**

You will first learn how to install and use PEAK, then apply it to selected benchmark cases and real applications on TACC systems.

Initial Training Goals: Learning PEAK

During the first stage of the project, you are expected to become familiar with the basic functionality of PEAK. This includes:

- Installing PEAK on the target HPC system.
- Running PEAK on simple test codes.
- Generating PEAK profiling reports.
- Understanding the information provided in the profiling reports, such as:
 - Which functions were captured.
 - How often each function was called.
 - The relative runtime contribution of different functions.
 - How profiling data can be interpreted in the context of application performance.

The purpose of this stage is to build enough familiarity with PEAK so that you can confidently apply it to more complex scientific and AI/ML workloads later in the project.

Research Direction 1: Math Libraries in Scientific Applications

The first research direction focuses on applying PEAK to scientific applications that are likely to use BLAS, LAPACK, FFTW, or related math library functions.

Main Objectives

You will use the list of the top 100 applications used on TACC Frontera as a starting point to identify applications that may rely on common math libraries. The goal is to study how these libraries are used in different scientific domains and how important they are to the overall performance of each application.

Planned Tasks

You will:

- Review the top 100 Frontera applications and identify candidate applications that are likely to use BLAS, LAPACK, FFTW, or similar math libraries.
- Learn how to build, install, and run selected applications on Stampede3.
- Search for available sample inputs, tutorial cases, or benchmark cases for the selected applications.
- Run initial benchmark or sample cases on the system.
- Use PEAK to profile selected BLAS, LAPACK, and FFTW functions within these applications.
- Analyze the profiling results to understand:
 - Which math library functions are called.
 - How frequently they are called.
 - How much runtime they contribute.
 - Whether the use of these libraries differs across applications or scientific domains.

Expected Outcomes

By the end of this direction, you should be able to summarize how different scientific applications use math libraries and identify whether certain domains or application types rely more heavily on BLAS, LAPACK, FFTW, or other numerical kernels.

Research Direction 2: Math Libraries in AI/ML Workflows

The second research direction focuses on the role of math libraries in AI/ML algorithms and workflows. Although many AI/ML packages are written in high-level languages such as Python, their performance often depends on optimized lower-level libraries for matrix operations, decompositions, and transforms.

Main Objectives

You will investigate common AI/ML algorithms and workflows that are likely to make substantial use of BLAS, LAPACK, FFTW, or related math libraries. They will then prepare representative benchmark cases and use PEAK to study the library usage patterns.

Planned Tasks

You will:

- Conduct a literature and documentation review to identify AI/ML algorithms or workflows that commonly rely on BLAS, LAPACK, FFTW, or similar functions.
- Identify representative sample cases or benchmark workflows.
- Prepare and run these workflows on the HPC system.
- Use PEAK to profile relevant math library functions.
- Analyze the profiling results to understand:
 - Which math library functions are used in each workflow.
 - How important these functions are to the overall runtime.
 - Whether different AI/ML workflows show different library usage patterns.
 - How the use of math libraries in AI/ML workflows compares with their use in traditional scientific applications.

Expected Outcomes

By the end of this direction, you should be able to describe how math libraries contribute to the performance of selected AI/ML workflows and compare these patterns with those observed in scientific applications.